

A soft exoskeleton with hybrid-jamming towards hand rehabilitation

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Introduction

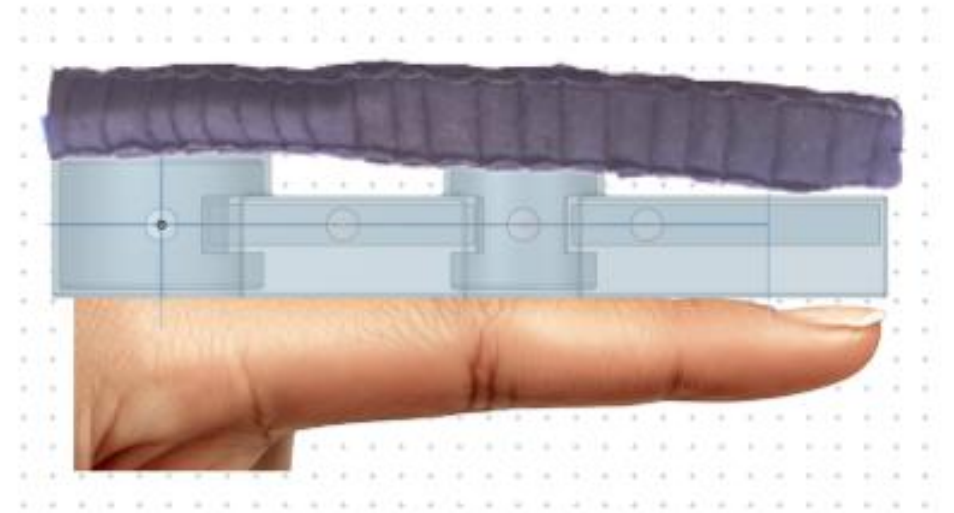
- Hand rehabilitation often requires repetitive, progressive, and variable movements with adjustable assistance (Reinkensmeyer et al., 2004).
- Soft exoskeleton gloves can provide clinical hand rehabilitation support (Rus and Tolley, 2015).
- This thesis aims to design, create, and test a bio-inspired, hybrid-jamming soft exoskeleton design that prioritizes graduated support and unique control for personalized rehabilitation.
- Hybrid jamming is the combination of granular jamming and laminar jamming. These refer to phenomena in which granular material or layers, respectively, placed in a flexible non-porous membrane can reversibly transition from a low stiffness state to a high stiffness state by applying a vacuum pressure.

David J. Reinkensmeyer et al. (2004). Robotic guidance and assistance as needed in neurorehabilitation. *Advanced Robotics*, 18(7), 695–715.

Daniela Rus & Michael T. Tolley (2015). Design, fabrication and control of soft robots. *Nature*, 521(7553), 467–475. <https://doi.org/10.1038/nature14543>

Finger-level design

- Pneumatic Fabric Actuator
 - Air-pressure determines bending force
- Hybrid Jamming Bio-Inspired Design
 - Granular jamming pockets act as joints (targeting MCP and DIP), laminar jamming pockets act as bones



Testing and Outcomes

- Will assess the feasibility of the design as a wearable exoskeleton, including overall device weight, ROM, actuation speed, stiffness enhancement, and maximum force output, with and without the jamming.
- The aim is to contribute data on the viability and limits of jamming mechanisms for enhancing soft robotic exoskeletons in the context of hand rehabilitation, as well as to provide insight into different control methods that could optimize cost, computation speed, and functionality for soft robotic exoskeletons.